

Week 1

Definition: Crime Scene

Crime Scene

- A crime scene is any **physical scene**, anywhere, that may provide potential evidence to an investigator. It may include a person's body, any type of building, vehicles, places in the open air or objects found at those locations.
- "Crime scene examination" therefore refers to an examination where forensic or scientific techniques are used to preserve and gather evidence of a crime.
- Reference: United Nations Office on Drugs and Crimes

Definition: Digital Evidence

Definition of Digital Evidence

- Digital Evidence includes **data on computer and mobile devices, including audio, video and image files as well as software and hardware**. Digital evidence can be a part of investigating most crimes, since material relevant to the crime may be recorded in digital form.
- Reference: National Institute of Standards and Technology (NIST)

Sources of Digital Evidence

Open Computer Systems	Embedded Computer Systems	Cloud Computing Systems	Non Volatile Storage Devices
• Servers • Laptops • Tablets	• Mobile Devices • IoT Devices	• AWS • Azure • Google Cloud	• SD cards • Hard Disks • Solid State Drives

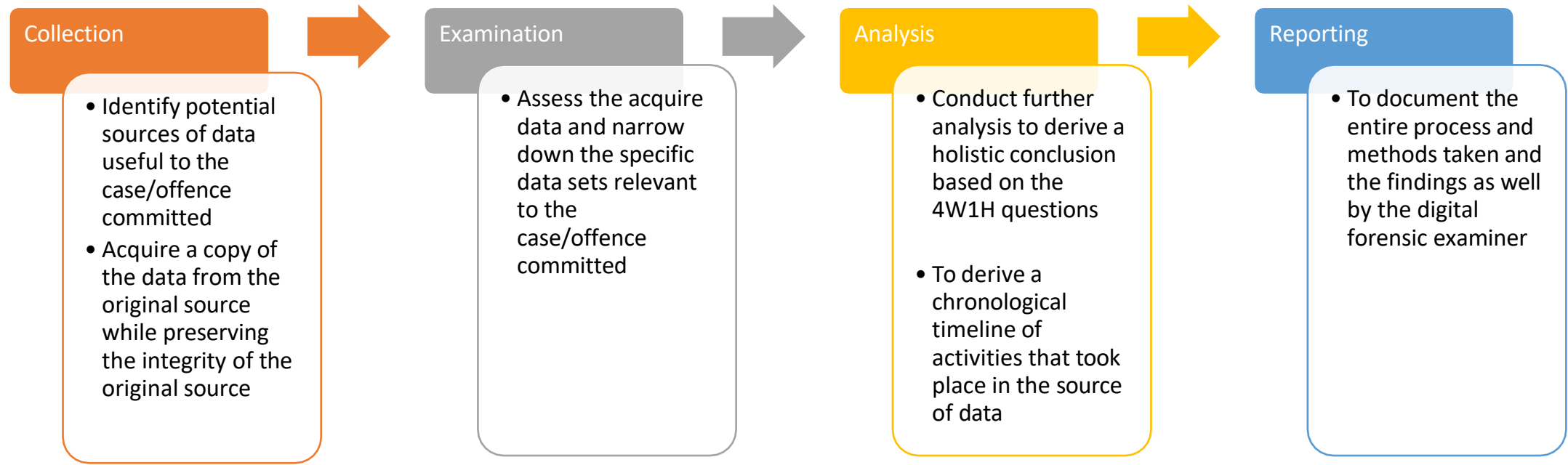
Digital Evidence as Alibi

- Digital evidence provides crucial information such as **time and location** for investigators to make sense of the chronological of events that took place associated to the crime.
- Examples of digital evidences that could provide information on time and location in mobile devices:
 - Timestamps of the messages and call logs exchanged between two parties
 - Metadata of media files (Photos, Videos, Audio Recordings)
 - GPS coordinates of recent visited locations

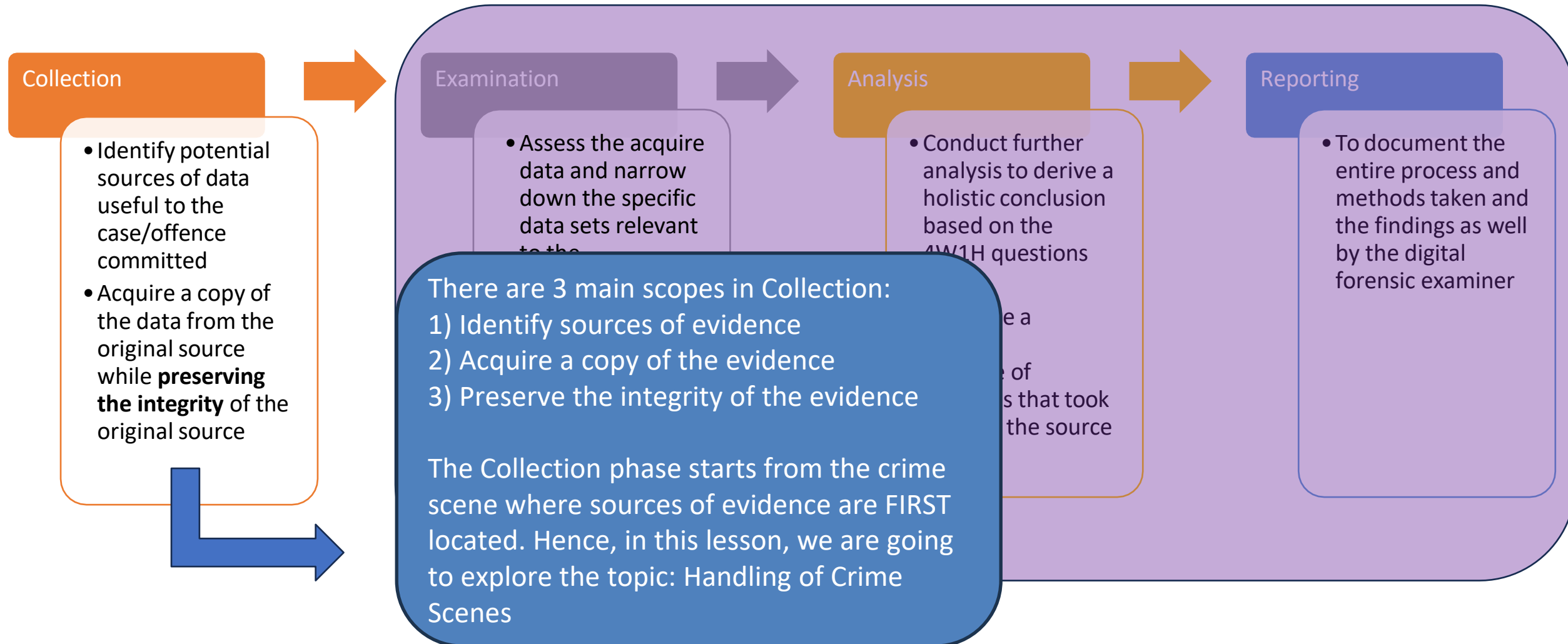
Digital Forensic Framework

Digital Forensic Framework by NIST

- The Digital Forensic Framework is designed to guide investigators in the process of handling digital evidence from end to end investigation. It comprises 4 main phases:



Digital Forensic Framework by NIST



Rules of Evidence

The Significance of Mobile Forensics in Criminal Cases

- Today, mobile devices are now considered as invaluable evidence to law enforcement agencies in solving criminal cases.
- The wealth of data a mobile device can provide (Photos, Videos, last visited locations, social media uploads) can contain critical data for criminal investigations.
- When conducting raids, **mobile devices are commonly found in crime scenes.**
- Hence, why there is a **need to establish a framework in handling mobile devices in investigations so that they can be admissible evidence to courts.**

Rules of Evidence

- In this day, investigations rely more on the information inside a mobile device as vital evidence. To prevail evidence in court requires an understanding of the rules of evidence. There are 5 general rules need to be followed to ensure that evidence to be useful in courts.
- Admissible, Authentic, Complete, Reliable and Believable

Rules of Evidence

Admissible	Authentic	Complete	Reliable	Believable
Evidence found at the crime scene must be preserve in a forensically sound manner. This is to ensure evidence validity when submitted to court.	The evidence must be tied to the incident in a relevant way to support the case.	When the evidence is presented in court, it must be clear and compete and reflect the entire incident.	Evidence acquired from the device must be reliable. Forensic tools and/or forensically sound methods are used to acquire the evidence.	A DF examiner must be able to explain with clarity and conciseness, what processes they used and how they preserved the integrity of the evidence.

Locard's Exchange Principle

Locard's Exchange Principle

- The main objective of an investigation is to **discover the truth**. Based on the evidence left on a crime scene, investigators must follow and tie them to the criminals and victims involved.
- According to Locard, a director of a first crime laboratory, a criminal will always **leave traces of his doings at the crime scene**. A criminal can leave all sorts of traces: Fingerprints, Blood Stains, Hair etc.

Locard's Exchange Principle

- And in the case of digital investigations, a criminal may leave traces related to digital evidence such as:
 - Computer and Mobile devices, External Storage devices, Emails, Audio Recordings
- Even though the criminal may have fled after he committed the crime, the traces left at the crime scene will always prevail. Hence, this phrase is commonly associated with Locard's Exchange Principle"
 - *"With contact between two items, there will be an exchange"*

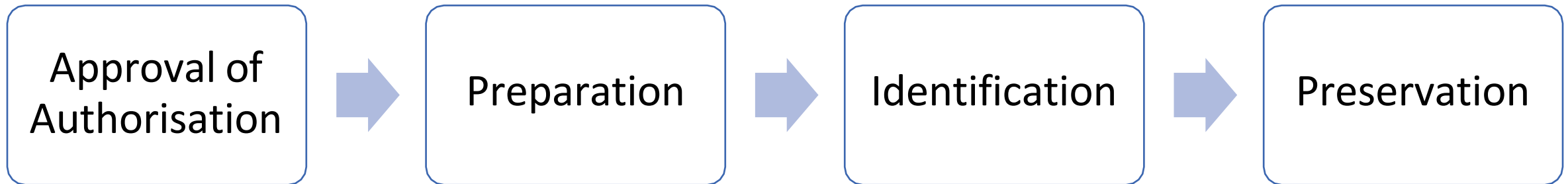
Locard's Exchange Principle

- With this concept in mind, we know understand why investigators conduct searches at the crime scenes. The role of every investigator is to uncover the traces left at the locations where the crime is committed.
- These traces may be deemed worthy of the case and are subjected to be used as digital evidence. Hence, the next chapter discuss about **Handling of Crime Scenes**.

Handling of Crime Scenes

Handling of Crime Scenes

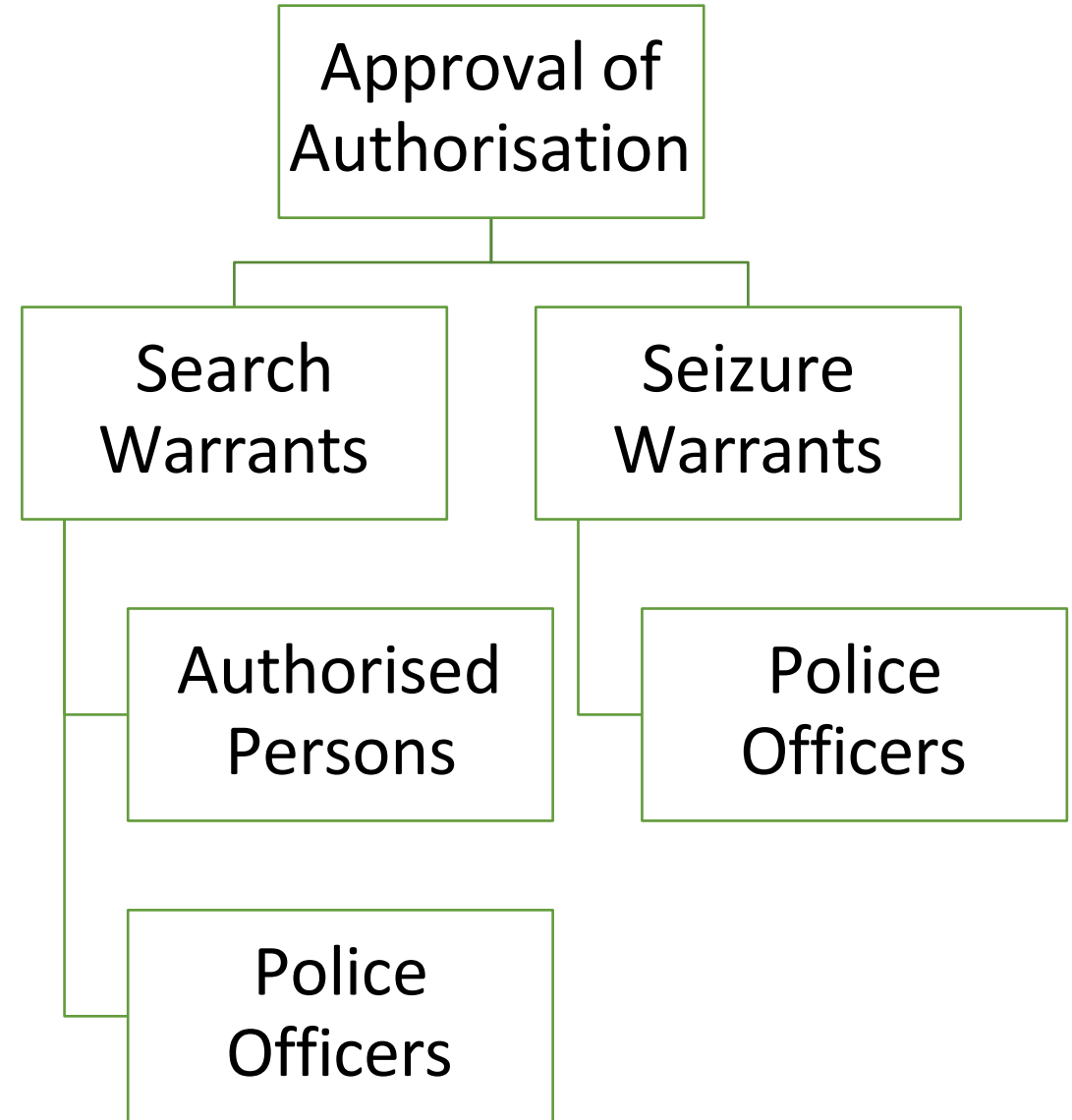
- Handling of crime scenes must be done meticulously to ensure the integrity of evidence remains intact and adhere to the Rules of Evidence. This framework entails the activities to be taken in handling of crime scenes.



Handling of Crime Scenes: Approval of Authorisation

Approval of Authorisation

- Before an investigation raid can commence, Police Officer and Authorised People need to seek two types of warrants. These warrants are only applicable to **criminal cases**.
 - Search Warrant
 - Seizure Warrant
- Digital Forensic (DF) examiners who attend crime scenes are considered as **Authorised Persons**.
- Search warrants can be authorised to both Police Officers and Authorised Persons.
- Seizure Warrants can be authorised to Police Officers only.



Introduction to Criminal Procedure Code 2010

Criminal Procedure Code 2010  

Status: Current version
as at 12 Mar 2026 

2 Interpretation

Table of Contents
or prosecution, etc.

Part 4 INFORMATION TO POLICE AND POWERS OF INVESTIGATION

Division 1 — Duties of police officer on receiving information about offences

- ☐ 14 Information about offences received by police
- ☐ 15 Information about offences received by authorised persons
- ☐ 16 Procedure in non-arrestable cases
- ☐ 17 Procedure when arrestable offence is suspected
- ☐ 18 Investigation in arrestable cases
- ☐ 19 Diary of proceedings in investigation
- ☐ 20 Power to order production of any document or other thing

Timeline ▾ Subsidiary Legislation ☒ Amendment Annotation

Actions ▾   

to commit, such an offence;

“complaint” means any allegation made orally or in writing to a Magistrate with a view to the Magistrate taking action under this Code that some person, whether known or unknown, has committed or is guilty of an offence;

“computer” has the meaning given by the Computer Misuse Act 1993;

“court” means the Court of Appeal, the General Division of the High Court, a Family Court, a Youth Court, a District Court or a Magistrate’s Court (as the case may be) which exercises criminal jurisdiction;

“Criminal Procedure Rules” —

- (a) means the Criminal Procedure Rules 2018 made under this Code and any other written law by the Criminal Procedure Rules Committee constituted under section 428A; and
- (b) includes any subsidiary legislation deemed under section 428A to be made under this Code.

“criminal record” means the record of any —

- (a) conviction in any court, or subordinate military court established under the Singapore Armed Forces Act 1972;

The CPC delineates the authority given to **police officers, authorised persons, public prosecutors and courts** in scope of work such as investigations, search and seizure, arrest and bails.

Search Warrants

Power to access computer

39.—(1) Subject to subsection (1A), a police officer or an authorised person investigating an arrestable offence may, at any time —

(a) access, inspect and check the operation in or from Singapore of a computer (whether in Singapore or elsewhere) that the police officer or authorised person has reasonable cause to suspect is or has been used in connection with, or contains or contained evidence relating to, the arrestable offence;

(b) use any such computer in or from Singapore, or cause any such computer to be used in or from Singapore —

(i) to search any data contained in or available to such computer; and

(ii) to make a copy of any such data;

(c) prevent any other person from gaining access to, or using, any such computer (including by changing any username, password or other authentication information required to gain access to the computer); or

(d) order any person —

(i) to stop accessing or using or to not access or use any such computer; or

(ii) to access or use any such computer only under such conditions as the police officer or authorised person may specify.

[19/2018]

[Act 5 of 2024 wef 01/08/2024]

In Section 39(1)(b) of CPC 2010, it is stated that **police officers and authorised people are allowed to:**

- 1) Have access to computer devices if they deem the devices to be used in the arrestable offence
- 2) Make a copy of the content from these devices
- 3) Prevent others from gaining access to these devices

Search Warrants

Power to access computer

39.—(1) Subject to subsection (1A), a police officer or an authorised person investigating an arrestable offence may, at any time —

- (a) access, inspect and check the operation in or from Singapore of a computer (whether in Singapore or elsewhere) that the police officer or authorised person has reasonable cause to suspect is or has been used in connection with, or contains or contained evidence relating to, the arrestable offence;
- (b) use any such computer in or from Singapore, or cause any such computer to be used in or from Singapore —
 - (i) to search any data contained in or available to such computer; and
 - (ii) to make a copy of any such data;

(c) prevent any other person from gaining access to, or using, any such computer (including by changing any username, password or other authentication information required to gain access to the computer); or

(d) order any person —

- (i) to stop accessing or using or to not access or use any such computer; or
- (ii) to access or use any such computer only under such conditions as the police officer or authorised person may specify.

[19/2018]

[Act 5 of 2024 wef 01/08/2024]

4) In most actual raids, suspects/alleged criminals are seen to panic and tries to wipe data in their computers.

This clause allows police officers to stop them from doing so. Therefore, the availability of data is not compromised.

Seizure Warrant (Police Officer)

Powers to seize property in certain circumstances

35.—(1) A police officer may seize, or prohibit the disposal of or dealing in, any property —

- (a) in respect of which an offence is suspected to have been committed;
- (b) which is suspected to have been used or intended to be used to commit an offence; or
- (c) which is suspected to constitute evidence of an offence.

(35)(1)(b) “..Used or intended to be used to commit an offence..”

Example: 2016 O’level Cheating case

Mobile phones, Bluetooth enabled earsets and desktops are considered as properties used to commit offence

(35)(1)(c) “..constitute evidence of an offence..”

Example: A woman receives messages of harassment from a stalker. Stalker has deleted all messages in his phone.

If the victim keeps the disturbing messages from the stalker, her mobile can be used as evidence under this clause.

Anton Piller Order

- Anton Piller Order (APO) is a combined search and seizure order warranted by court applicable to **civil cases** only.
- Examples of civil cases:
 - Breach of Contracts
 - Disputes between companies
 - Disputes between neighbours
 - An employee receives mistreatment from his employer

Interpol Notices

- Singapore is one of the member countries associated with Interpol.
- In heavy duty cases where our police force requires international help in information regarding such as whereabouts of a runaway offender or unidentified bodies, SPF may make requests to Interpol to issue notices to other member countries to share crime related information.

Interpol Notices

Type of Notice	Objective
Red	To seek location and arrest of a person wanted for prosecution or to serve a sentence.
Yellow	To help locate missing persons, often minors, or to help identify persons who are unable to identify themselves.
Blue	To collect additional information about a person's identity, location or activities in relation to a criminal investigation.
Black	To seek information on unidentified bodies
Green	To provide warning about a person's criminal activities, where the person is considered to be a possible threat to public safety
Orange	To warn of an event, a person, an object or process representing a serious and imminent threat to public safety.
Purple	To seek or provide information on modus operandi, objects, devices and concealment methods used by criminals.

Handling of Crime Scenes: Preparation

Preparation

- After receiving the search and seizure warrants, investigators must curate a plan in ensuring the handling of digital evidence a crime scene adheres to the Rules of Evidence.
- In this phase, DF examiners are to prepare a list of necessary items to collect the evidences at crime scene such as:
 - Hardware Equipment
 - Mobile Acquisition Software
 - Storage Media
 - Documentation (Example: Chain of custody)

Chain of Custody

The Chain of Custody important in every criminal case as it comply with the first Rule of Evidence: **Admissible**.

The document tracks information like:

- 1) The condition of the physical evidence
- 2) The people involved in handling the physical evidence.
- 3) Date and Time of every single evidence handover.

Property Record Number: _____				
Anywhere Police Department EVIDENCE CHAIN OF CUSTODY TRACKING FORM				
Case Number: _____ Offense: _____				
Submitting Officer: (Name/ID#) _____				
Victim: _____				
Suspect: _____				
Date/Time Seized: _____ Location of Seizure: _____				
Description of Evidence				
Item #	Quantity	Description of Item (Model, Serial #, Condition, Marks, Scratches)		
Chain of Custody				
Item #	Date/Time	Released by (Signature & ID#)	Received by (Signature & ID#)	Comments/Location

Handling of Crime Scenes: Identification

Identification

- Upon reaching the location of the crime, A DF examiner may spot several electronic devices. However, not all devices are potential sources of evidence.
- Surveying is to be done to determine the electronic devices which could be potentially used as evidence. Surveying activities includes:
 - 1st Priority: Capture photos of devices that are running applications (Example: Ongoing downloads).
 - 2nd Priority: Capture photos of devices that are already in powered on state.
 - 3rd Priority: Delegate the questioning to Police Officer to interview the suspect (Which devices are frequently accessed by the suspect?. DF examiners are not allowed to conduct interviews.
- After identifying the few electronic devices of interest, we move on to the last stage of the Handling of Crime Scenes: Preservation **(to be covered in the next session)**.